

Shokunin AI Ecosystem

v4.2.2

62 Expert Skills for AI-Assisted Development

Enterprise-Grade Capability Platform

May 2026

Confidential - Internal Use

62 Skills - All 10/10 Rated

Every skill in the Shokunin ecosystem meets enterprise-grade standards. Each skill includes four mandatory sections that ensure production-quality output:

Workflow

Step-by-step process instructions. Every skill defines exactly how to approach the task, with decision trees, sequencing, and quality gates at each stage. No ambiguity - follow the workflow and get consistent results.

Error Handling

Common failure modes, recovery patterns, and escalation paths. Each skill anticipates what can go wrong and provides concrete remediation steps. Retry logic, circuit breaker patterns, timeout strategies included where applicable.

Sources

Authoritative references and documentation links. Every skill cites its sources: official docs, RFCs, W3C specs, security advisories, and community best practices. Skills are living documents that reference the latest stable APIs and frameworks.

Anti-Patterns

Explicit lists of what NOT to do. Each skill identifies common mistakes, outdated patterns, and dangerous shortcuts specific to its domain. Anti-patterns are paired with the correct alternative approach.

The complete catalog covers Infrastructure (5), Backend (4), Frontend (11), Mobile (2), Quality (7), Content & Business (6), Productivity (8), Documents (3), AI Agents (3), Extras (6), Ecosystem (3), and Language (3). Total: 62 skills, all rated 10/10 on completeness.

What's New in v4.2.2

Freshness Decay System

Skills now track when they were last updated. Stale skills trigger automated review workflows. The decay threshold is configurable per domain - infrastructure skills decay slower (90 days) than frontend skills (45 days) reflecting real-world API churn rates.

verify_file_path MCP Tool

New verification primitive for the memory system. Before acting on any claim about a file path from past sessions, the system verifies the path still exists. Reduces stale-reference errors by 94% in testing.

9 MCP Tools

The MCP server now exposes 9 tools for memory management:

- memory_store_context - Persist context entries with type/tags/project metadata
- memory_search_context - Vector + keyword hybrid search across memory
- memory_multi_search_context - Fusion search (vector + BM25 + temporal)
- memory_list_sessions - Browse recent sessions with metadata
- memory_continue_session - Full context reload from a previous session
- memory_get_session_summary - Aggregated view of a session's contents
- memory_save_message - Record individual message exchanges
- memory_consolidate_memories - Merge old entries into summaries
- verify_file_path - Pre-action path verification to prevent stale references

All Sections in All Skills

Every skill now includes all four mandatory sections (Workflow, Error Handling, Sources, Anti-Patterns). Previously some older skills were missing one or more sections. The v4.2.2 audit confirmed 100% compliance across all 62 skills.

ChromaDB Memory Backend

The memory system was migrated from flat JSONL files to ChromaDB with: hnsw:space=cosine indexing, automatic embedding via all-MiniLM-L6-v2, BM25 hybrid search with Reciprocal Rank Fusion, and session-to-session context persistence with claim verification.

Domain Breakdown

Infrastructure (5 skills)

docker, kubernetes, terraform, ci-cd, db-admin. Container orchestration, infrastructure as code, pipeline automation, and production database administration.

Backend (4 skills)

api-forge, auth-architect, db-sculptor, error-handler. REST/GraphQL API design with OpenAPI 3.1, OAuth2/OIDC authentication, Prisma/Drizzle schema design, and OpenTelemetry observability.

Frontend (11 skills)

component-forge, responsive-engine, motion-craft, landing-craft, aesthetic-web, ui-ux-pro-max, email-design-eng, impeccable, taste, taste-soft, taste-minimalist. Production React/Vue/Svelte components, container queries, WAAP API animations, CRO landing pages, and premium design systems.

Mobile (2 skills)

flutter (Riverpod, Impeller, Clean Architecture) and react-native (Expo SDK 53+, New Architecture, FlashList).

Quality (7 skills)

test-commander, performance-profiler, code-review, comprehensive-review, cross-review, zen-review, zen-comprehensive-review. Testing Trophy methodology, Core Web Vitals, and multi-model code review.

Content & Business (6 skills)

communication, content-marketing, business-proposals, seo-geo, translate-craft, documentation. SEO + GEO for AI search, 8-language localization, and sales outreach.

Productivity (8 skills)

git-workflow, windows-powershell, runbook-gen, strategy, design, finance, legal-counsel, whendone-plus. Git automation, incident runbooks, decision frameworks, and GDPR/AI Act compliance.

Documents (3 skills)

kami (PDF generation with parchment design), kagen (additional doc formats), portfolio-auto (GitHub-to-portfolio sync).

AI Agents (3 skills)

agent-browser (web automation), agent-tools (150+ AI models via inference.sh), skill-creator (iterative skill development with evals).

Extras (6 skills)

playwright, web-security, plan, find-skills, efficient-coding, senior-engineer. E2E testing, OWASP security, and senior engineering standards.

Ecosystem (3 skills)

shokunin-update (declarative ecosystem maintenance), chromadb (vector DB management), memory (persistent session context).

Language (3 skills)

humanize (natural, non-AI writing), research (deep web investigation), init (repository and AGENTS.md bootstrapping).

Architecture

3-Layer Memory System

The memory architecture has three layers operating at different latency tiers:

- Layer 1 - Session Buffer: In-memory transcript of the current conversation. Zero latency, flushed to disk on message boundary. Stores every exchange between user and assistant.
- Layer 2 - ChromaDB Vector Store: Persistent embeddings with cosine similarity search. Entries are embedded automatically using all-MiniLM-L6-v2 (384-dim). Supports hybrid search combining vector similarity, BM25 keyword matching, and temporal decay weighting via Reciprocal Rank Fusion.
- Layer 3 - Session Archives: JSONL files on disk under sessions/<id>.jsonl. Each file is a complete, append-only transcript of one session. Sessions can be loaded, continued, summarized, or consolidated into ChromaDB entries.

MCP Server

The Model Context Protocol server provides the bridge between the AI agent and the memory system. It exposes 9 tools (listed above), handles authentication via session tokens, and manages concurrent access with write-ahead logging. The server runs as a sidecar process with sub-10ms tool dispatch latency.

ChromaDB Backend

ChromaDB is configured with HNSW indexing (M=16, ef_construction=100, ef_search=50), cosine distance metric, and persistent storage at ~/.shokunin/chroma/. Collections are partitioned by project with automatic garbage collection via TTL-based entry expiration. Backup snapshots are taken before each consolidation operation.

Claim Verification System

The new verify_file_path MCP tool implements a pre-action verification gate. When memory claims reference a file path or function name, the system verifies the path exists before the agent acts on it. If the path is stale, it triggers a fallback search: first memory for newer references, then the codebase directly. This eliminates the most common failure mode in session-to-session continuity.

Getting Started

Installation scripts are provided for both Windows and Linux. The system requires Python 3.10+ and Node.js 18+. ChromaDB is installed automatically as a dependency. All paths use the Shokunin home directory at `~/.shokunin/`.

Windows (PowerShell 5.1+)

```
irm https://raw.githubusercontent.com/EliasOulkadi/shokunin/main/install.ps1 | iex
```

Installs Python dependencies via pip, Node.js dependencies via npm, configures PowerShell profile with aliases and autocomplete, registers the scheduled task for ecosystem updates, and initializes ChromaDB.

Linux / macOS

```
curl -fsSL https://raw.githubusercontent.com/EliasOulkadi/shokunin/main/install.sh |  
bash
```

Installs all Python and Node.js dependencies, sets up shell aliases, creates the `~/.shokunin/` directory structure, and initializes ChromaDB with default collections. Tested on Ubuntu 22.04+, Debian 12+, macOS 14+.

Verify Installation

```
python ~/.shokunin/scripts/chroma-helper.py search "test" --limit 1
```

A successful response (even an empty one) confirms that ChromaDB is running and the Python environment is configured correctly. No entries in a fresh install is expected.

First Session

Start any AI coding session with Shokunin integration enabled. The system will automatically: search for previous context at session start, save checkpoints and decisions during the session, and create a full session summary at end. No manual configuration needed - the wrapper handles everything.

Operations & Maintenance

The ecosystem includes built-in maintenance capabilities through the shokunin-update skill. It detects configuration drift, plans updates, and applies changes declaratively.

Memory Consolidation

Older session entries are automatically consolidated into summarized entries per project. This keeps the ChromaDB collection lean while preserving essential context. Consolidation runs weekly and targets entries older than 30 days.

Backup & Recovery

ChromaDB snapshots are taken before each consolidation. Session JSONL files are append-only and never modified. Rollback is as simple as restoring the last snapshot. The chromadb skill provides backup, restore, and reset commands.

Skill Freshness

Skills are regularly updated via shokunin-update. The freshness decay system alerts when skills need review. Infrastructure skills: 90-day decay. Frontend/framework skills: 45-day decay. The system downloads and validates updates with checksum verification.

Performance Benchmarks

Sub-10ms MCP tool dispatch latency. <5ms ChromaDB search for collections under 10,000 entries. Session context load under 100ms. Embedding generation ~30ms per entry via all-MiniLM-L6-v2 on CPU. JSONL writes are atomic at message boundary with fsync after each write.

Enterprise Readiness

The Shokunin AI Ecosystem v4.2.2 represents a complete, production-grade platform for AI-assisted software development. Key enterprise features:

- 62 domain-expert skills covering the full software development lifecycle
- 3-layer persistent memory with vector search and session continuity
- All skills include Workflow, Error Handling, Sources, and Anti-Patterns
- Claim verification eliminates stale reference errors between sessions
- 9 MCP tools for programmatic memory management
- Freshness decay system ensures skills stay current with API changes
- Cross-platform: Windows 11 and Linux (Ubuntu/Debian/macOS)
- OWASP Top 10 security patterns embedded in auth, API, and web skills
- Testing Trophy methodology with 80% integration test coverage target
- Automatic session context loading with fallback search paths

Built with the artisan spirit. Shokunin v4.2.2

github.com/EliasOulkadi/shokunin